An Optimized Deep Learning-Based Structure-from-Motion Pipeline for Efficient Glacier 3D Reconstruction

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Abstract. Glacier three-dimensional (3D) reconstruction remains challenging for traditional Structure-from-Motion (SfM) pipelines under low-texture and reflective conditions. We propose an optimized deep learning-based pipeline using SuperPoint and LightGlue, accelerated with ONNX Runtime. Experiments on UAV glacier imagery show sub-pixel reprojection error, denser point clouds, and reduced GPU memory usage compared to COLMAP and PyTorch-native baselines. This highlights the efficiency and scalability of deep learning approaches for large-scale glacier reconstruction.

Keywords: Glacier Reconstruction · Structure-from-Motion · Superpoint · LightGlue

1 Introduction

Glacier 3D reconstruction [1–3] is of critical importance for advancing the understanding of climate change, monitoring variations in glacier volume, and analyzing ice dynamics. High-fidelity 3D models facilitate long-term observation of glacier retreat, mass balance fluctuations, and surface motion, thereby providing indispensable insights for climate science and natural hazard assessment [4–6].

Traditional SfM pipelines [3, 7], particularly those employing scale-invariant feature transform (SIFT) [8, 9] for feature extraction and matching [10, 11], frequently encounter severe limitations in glacier environments. The prevalence of low-texture surfaces, intense specular reflections, and large-baseline multi-view imagery substantially undermines the robustness and accuracy of such traditional approaches, often resulting in incomplete reconstructions or geometric distortions [12, 13].

In this paper, we present *An Optimized Deep Learning-Based SfM Pipeline for Efficient Glacier 3D Reconstruction*. The contributions of this work can be summarized as follows:

- We propose an optimized deep learning-based SfM pipeline [14, 15] that substitutes the traditional SIFT–COLMAP framework with SuperPoint [16] for

feature extraction and LightGlue [17] for feature matching, while leveraging ONNX Runtime [18] with FP16(16-bit floating point) inference for efficiency.
- We evaluate the proposed pipeline on glacier datasets, demonstrating improvements in both reconstruction efficiency and scalability when compared to conventional methods.
- We conduct ablation experiments contrasting the PyTorch-native [19] SuperPoint–LightGlue pipeline with its ONNX Runtime FP16-optimized counterpart, systematically analyzing runtime performance and memory utilization.

2 Related Work

2.1 Traditional Feature Extraction Methods

One of the most widely used traditional computer vision methods in SfM is SIFT, which detects keypoints by locating local extrema in a Difference-of-Gaussians (DoG) pyramid [8] across multiple scales. This approach provides robustness to variations in both scale and rotation. Each keypoint is then assigned an orientation based on local image gradients, allowing invariance to rotation [20, 21]. Finally, SIFT constructs a distinctive 128-dimensional descriptor by summarizing gradient orientations within a local neighborhood around the keypoint. This combination of scale-space detection, orientation assignment, and gradient-based descriptors has made SIFT a cornerstone of classical SfM pipelines. Nevertheless, in glacier imagery with repetitive patterns, low texture, and strong reflections, SIFT often produces sparse or unreliable matches, limiting its effectiveness even when GPU acceleration is applied.

2.2 Deep Learning-Based Feature Extraction and Matching

Recent advances in deep learning have enabled learned alternatives to traditional descriptors. SuperPoint, for example, is a self-supervised convolutional neural network [22] that jointly learns interest point detection and descriptor extraction. It produces repeatable keypoints and compact descriptors that are more robust in texture-poor or noisy environments compared to SIFT. For feature matching, LightGlue introduces a transformer-based architecture [23] that dynamically decides how many iterations are required to establish correspondences between two sets of features. Unlike static matching strategies, LightGlue adapts to the difficulty of image pairs, making it efficient for easy cases while maintaining accuracy in challenging ones. This flexibility reduces computation without sacrificing robustness, allowing large-scale SfM tasks to scale more effectively [24]. Building on these developments, our work proposes an optimized deep learning-based SfM pipeline, with a particular focus on accelerating feature extraction and matching for glacier reconstruction through FP16 inference and ONNX Runtime optimization.

3 Methodology

3.1 Pipeline Overview

This work introduces a optimized SfM pipeline in which deep learning-based feature extraction and matching replace the conventional SIFT descriptors employed in the standard COLMAP framework. The new design preserves the classical SfM geometry estimation stages while redefining the feature processing component. In the SIFT-COLMAP pipeline, reconstruction relies on SIFT-based descriptors and exhaustive matching, followed by pose estimation, outlier rejection, and bundle adjustment. While effective, SIFT features often degrade in glacier imagery due to low contrast and repetitive patterns. In contrast, the proposed pipeline incorporates SuperPoint for keypoint detection and description, combined with LightGlue for transformer-based feature matching. The subsequent SfM steps (PnP, RANSAC, BA) remain unchanged, enabling a direct comparison between the classical and deep learning-based pipelines. The implementation of the proposed pipeline is available here: `https://github.com/AppleJack9050/glacier_pipeline`.

3.2 Deep Learning-Based Feature Extraction and Matching

The front end of the pipeline is implemented using **SuperPoint**, a self-supervised network for joint keypoint detection and description. Compared with SIFT, SuperPoint offers features that are more robust to scale, rotation, and contrast variations, which are critical in glacial environments. Feature correspondences are then established using **LightGlue**, a lightweight transformer-based matcher that dynamically adjusts its computational load depending on image complexity. In glacier imagery, where homogeneous surfaces are common, we configure the system to retain up to $N = 4096$ keypoints per image, with a non-maximum suppression (NMS) radius [25, 26] to reduce redundancy while ensuring spatial coverage.

3.3 Optimization Strategy

To achieve efficiency on large-scale glacier datasets, both SuperPoint and LightGlue are exported to the ONNX format and executed through ONNX Runtime with half-precision (FP16) inference. This optimization reduces computational overhead and memory usage while maintaining numerical stability relative to the PyTorch baseline. By integrating ONNX Runtime acceleration, the proposed system is capable of processing significantly larger datasets under constrained hardware resources.

3.4 Integration with SfM

The proposed front end is integrated into the **Hierarchical Localization (HLOC)** framework [24, 27], which provides a flexible interface for combining learned

feature extractors and matchers with traditional SfM backends. In this work, SuperPoint and LightGlue are coupled within HLOC to produce feature correspondences, which are then passed to COLMAP for geometric reasoning. Camera poses are estimated via PnP, outliers are rejected through RANSAC, and bundle adjustment is employed for global refinement. This design ensures that efficiency improvements stem from the optimized feature extraction and matching stages, while the SfM backend remains identical to the SIFT-COLMAP baseline.

3.5 Ablation Study Setup

The ablation study compares two configurations: (1) **PyTorch Baseline**, where SuperPoint and LightGlue are executed natively in PyTorch with full precision; and (2) **Optimized Inference**, where the same models are converted to ONNX Runtime with FP16 acceleration. The comparison focuses exclusively on the feature extraction and feature matching stages, with evaluation metrics including average inference time and peak GPU memory usage. This setup isolates the contribution of the proposed optimization to computational efficiency, independent of the geometric SfM backend.

4 Experiment

4.1 Dataset

We use a UAV glacier dataset of 567 images captured by a DJI FC330 camera, modeled as a *pinhole camera* with distortion parameters. The native resolution is 2000×1500 pixels, corresponding to a ground sampling distance of 3.49 mm/pixel at an altitude of 11.6 m, covering about 1.92×10^3 m^2. All images were undistorted and downscaled to a maximum of 2000×1498 pixels to reduce computation while preserving geometry. The dataset is publicly available at the UNAVCO TLS repository.

4.2 Experimental Setup

All experiments were conducted under a unified SfM configuration to ensure comparability. The hardware environment consisted of an Intel Core i7-9700X CPU, 48 GB of RAM, and an NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM, running Ubuntu 22.04 LTS. For the traditional baseline, we utilized the SIFT-based COLMAP pipeline. Since SIFT implementations exist for both CPU and GPU, we adopted the GPU-accelerated version to reduce runtime overhead and provide a fairer comparison against the deep learning-based pipelines. On the learning-based side, the baseline feature extraction and matching were implemented in PyTorch, while optimized inference was performed with ONNX Runtime in FP16 mode. Thus, three pipelines were evaluated: the GPU-accelerated SIFT-based COLMAP system, the PyTorch-native SuperPoint–LightGlue pipeline, and the ONNX Runtime FP16 optimized variant.

4.3 Evaluation Metrics

The evaluation focused on two categories of metrics:

- **Reconstruction Quality:** Root Mean Square Reprojection Error (RMSE) [28, 29] and the number of 3D points were used as quantitative measures of reconstruction accuracy.
- **Reconstruction Efficiency:** We reported three indicators to assess computational performance: overall reconstruction speed, average GPU utilization, and peak Videa RAM (VRAM) consumption.

4.4 Results

Traditional vs. Optimized Pipeline Relative to a standard COLMAP baseline, the optimized pipeline maintains sub-pixel RMSE and improves geometric completeness: the final point cloud contains more 3D points with broader spatial coverage, while the number of registered images remains comparable (Table 1). All results are averaged over 10 independent runs to ensure statistical reliability.

Table 1: Comparison of Traditional SIFT and Optimized Pipeline. All values are averaged over 10 runs; the $\pm$ indicates one standard deviation.

Method	**RMSE (px)**	**3D Points**
Traditional SIFT	0.5939 ± 0.018	$482,213 \pm 187$
Optimized Pipeline	0.4936 ± 0.015	$530,434 \pm 243$

Ablation: PyTorch Native vs. ONNX Runtime (FP16) Replacing the PyTorch-native feature extraction and matching with ONNX Runtime FP16 reduces the mean time for these stages to 4.19 ± 0.33 minutes and peak GPU memory to 15.5 ± 0.4 GB (Table 2), alleviating the primary throughput bottleneck. Each configuration was tested across 10 runs, and the reported values represent the mean $\pm$ standard deviation.

Table 2: PyTorch Native vs. ONNX Runtime FP16 on Feature Extraction and Matching. All values are averaged over 10 runs; the $\pm$ indicates one standard deviation.

Method	**Runtime (mins)**	**GPU Memory (GB)**
PyTorch Native (FP32)	5.86 ± 0.47	19.1 ± 0.5
ONNX Runtime FP16	4.19 ± 0.33	15.5 ± 0.4

Qualitative Visualization The baseline panel is a densified Multi-View Stereo(MVS) [30] reconstruction, whereas our results are sparse SfM point clouds. Within the sparse regime, structure is comparable to the baseline, with at most slight densification in high-gradient areas and no observable increase in artifacts (Figure 1).

(a) Ground truth model (b) COLMAP-based model (c) DL-based model

Fig. 1: Snapshots of three reconstruction models. The first is the official ground truth reconstruction, the second is obtained using the SuperPoint+SuperGlue framework, and the third is generated with COLMAP.

5 Discussion

Accuracy and completeness. The optimized pipeline delivers lower reprojection error while simultaneously increasing the number of reconstructed points (Table 1). This indicates that the use of learned features coupled with transformer-based match filtering enhances correspondence recall without compromising geometric consistency. The improvements are particularly visible in regions of high surface gradient, such as crevasse edges or steep ice walls, where traditional SIFT features tend to fail due to low contrast and repeated patterns. By leveraging SuperPoint's scale- and contrast-invariant descriptors together with LightGlue's context-aware filtering, the system is able to recover matches that extend coverage into previously underrepresented areas. As a result, the reconstructed point cloud not only becomes denser but also spatially more uniform, reducing holes in the geometry and enabling better support for downstream dense stereo stages.

Efficiency and compute. Replacing PyTorch-native inference with ONNX Runtime FP16 results in a measurable reduction in runtime and peak GPU memory during feature extraction and matching (Table 2), effectively alleviating what is often the dominant computational bottleneck in learning-based SfM pipelines. The runtime gains stem from both half-precision tensor operations and optimized graph execution, while reduced memory demand allows for larger batch sizes or processing of higher-resolution imagery within the same hardware constraints. Despite these improvements, end-to-end SfM performance remains constrained by components that are inherently sequential or CPU-bound, including incremental mapping, bundle adjustment, and large-scale I/O. These stages restrict

the pipeline from fully saturating GPU resources, even when powerful hardware, such as an NVIDIA RTX 4090, is available. This suggests that while front-end acceleration is crucial, further advances in the parallelization of back-end geometry estimation will be necessary to achieve true real-time reconstruction performance.

Qualitative interpretation. The qualitative comparison panel juxtaposes a densified MVS reconstruction against sparse SfM outputs (Figure 1). The apparent differences in density stem primarily from the algorithmic distinction between multi-view stereo (which explicitly aims for dense surface recovery) and structure-from-motion (which remains sparse by design). Within the sparse regime, however, the optimized pipeline demonstrates clear advantages. It consistently recovers geometry comparable to or slightly richer than the traditional baseline, especially in structurally complex areas such as heavily crevassed or fractured glacier surfaces. The learned features also exhibit improved repeatability across low-texture ice fields, reducing local dropout of points and producing more coherent coverage. Importantly, no spurious structures or visible artifacts were introduced, underscoring that the observed density gains do not compromise the overall fidelity of the recovered 3D model.

Limitations and scope. Several limitations bound the interpretation of these results. First, all images were downscaled prior to processing, which caps the maximum achievable accuracy and likely underestimates the benefits of learned features on native-resolution inputs where more fine-scale detail is preserved. Second, the evaluation is restricted to a single UAV/glacier dataset with a single camera model; while glaciers represent a challenging domain due to texture scarcity and repetitive patterns, generalization to other terrains (e.g., vegetation, urban facades), different seasons, or varying imaging geometries is not yet tested. Third, the accuracy metric used—root mean square error in pixel space—effectively measures reprojection consistency but does not directly quantify absolute 3D surface error. As such, conclusions about geodetic accuracy or volumetric change detection remain tentative. Addressing these limitations will require larger-scale, cross-domain evaluations and integration with ground-truth datasets or lidar references to assess the absolute quality of 3D reconstructions.

6 Conclusion

We presented an optimized deep learning-based SfM pipeline for glacier 3D reconstruction, combining SuperPoint and LightGlue with ONNX Runtime FP16 acceleration. The system achieves lower reprojection error, denser point clouds, and reduced GPU memory usage compared to SIFT-COLMAP and PyTorch baselines, while preserving structural fidelity. These results highlight the efficiency and scalability of learned features for challenging cryospheric environments.

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